TOAST Stroke Classification

Trial of Org 10172 in Acute Stroke Treatment (TOAST) diagnostic criteria for ischemic stroke etiology.

TOAST Diagnostic Classification for Ischemic Stroke Etiology (LAA, SVO, CE, Other, Undetermined)

1. Large-Artery Atherosclerosis (LAA)

- **Clinical:** Cortical signs (aphasia, neglect, gaze deviation) or brainstem/cerebellar syndrome.
- **Imaging:** Cortical or subcortical/cerebellar/brainstem infarct matching the vascular territory.
- **Vascular: > 50% stenosis** or occlusion of the relevant major extracranial (carotid, vertebral) or intracranial artery.
- **Exclusion:** Must exclude a high-risk cardioembolic source.

2. Small-Vessel Occlusion (SVO / Lacune)

- **Clinical:** Classic lacunar syndrome (pure motor, pure sensory, sensorimotor, ataxic hemiparesis, clumsy hand) **WITHOUT** cortical signs.
- **Imaging:** Normal scan or deep subcortical/brainstem lesion **≤ 2.0 cm**.
- **Vascular/Cardiac:** Relevant artery must lack >50% stenosis, and patient must lack high-risk cardioembolic sources.

4. Other Determined Etiology (ODE)

- **Clinical/Imaging:** Infarction of any size with diagnostic proof of a rare/specific underlying mechanism:
- Arterial dissection (e.g. carotid or vertebral dissection)
- CNS vasculitis or systemic vasculopathy
- RCVS (Reversible Cerebral Vasoconstriction Syndrome)
- Moya-Moya disease, CADASIL, or Fibromuscular Dysplasia
- Prothrombotic/hypercoagulable state (APLS, cancer, DIC)

3. Cardioembolism (CE)

HIGH-RISK SOURCES:

- Mechanical prosthetic valve
- Mitral stenosis w/ AFib
- Atrial fibrillation / flutter
- Left atrial / LAA thrombus
- Sick sinus syndrome
- Recent MI (< 4 weeks)
- LVEF < 28% or LV thrombus
- Infective endocarditis

MEDIUM-RISK SOURCES:

- Mitral valve prolapse
- Mitral ring calcification
- Mitral stenosis w/o AFib
- Left atrial turbulence/smoke
- PFO w/ atrial septal aneurysm
- Atrial flutter (isolated)
- Bioprosthetic heart valve
- Nonbacterial endocarditis

5. Undetermined Etiology (UDE / ESUS)

- **Two or more potential causes:** e.g. 60% carotid stenosis AND atrial fibrillation (unable to assign single primary cause).
- **Incomplete evaluation:** Imaging or cardiac workup pending/incomplete.
- **Cryptogenic / ESUS:** Comprehensive workup unrevealing (Embolic Stroke of Undetermined Source).

Mandatory Etiologic Workup Checklist

- ☑ **Parenchymal:** MRI Brain (DWI/ADC) preferred, or CT Head.
- ☑ **Vascular:** CTA or MRA Head & Neck (or Carotid Duplex + TCD).
- ☑ **Rhythm:** EKG + Continuous Telemetry ≥ 24h (or loop recorder).
- ☑ **Cardiac:** TTE required; consider TEE if cryptogenic / ESUS suspected.

WHY SUBTYPE MATTERS — 2026 SECONDARY PREVENTION

Etiology drives prevention: **cardioembolic** → oral anticoagulation; **non-cardioembolic** (large-artery, small-vessel, or undetermined/ESUS) → antiplatelet therapy ± short-course DAPT. ESUS trials (NAVIGATE-ESUS, RE-SPECT ESUS) found empiric DOAC anticoagulation no better than aspirin — so ESUS is treated with antiplatelets pending a defined source. See the DAPT and Aspirin Failure cards.

The factor XIa inhibitor class (new for 2026): these oral agents aim to uncouple antithrombotic efficacy from bleeding risk. Both phase-2 stroke trials were neutral — **PACIFIC-Stroke** (asundexian) and **AXIOMATIC-SSP** (milvexian, a covert-infarct-heavy composite). **OCEANIC-STROKE** (phase 3) was then the first positive trial of the class: adding oral **asundexian** to antiplatelet therapy in non-cardioembolic stroke/high-risk TIA cut recurrent ischemic stroke (HR 0.74) with no significant excess of major bleeding. The milvexian phase-3 **LIBREXIA-STROKE** is ongoing. Note the class does NOT replace guideline anticoagulation for AF (OCEANIC-AF stopped early for inferiority to apixaban).

Original Study: Adams HP Jr, et al. TOAST. *Stroke*. 1993;24:35-41. **PMID: 7678184**.
AHA/ASA Guideline: Kleindorfer DO, et al. 2021 Stroke Prevention. *Stroke*. 2021;52:e364-e467. **PMID: 34024117**.
Factor XIa class: OCEANIC-STROKE (Sharma M, et al. *N Engl J Med*. 2026;394:1467-1479. **PMID: 41985132**) • PACIFIC-Stroke (Shoamanesh A, et al. *Lancet*. 2022;400:997-1007. **PMID: 36063821**) • AXIOMATIC-SSP (Sharma M, et al. *Lancet Neurol*. 2024;23:46-59. **PMID: 38101902**).